Audio Classification - Cat or Dog?

Project Overview

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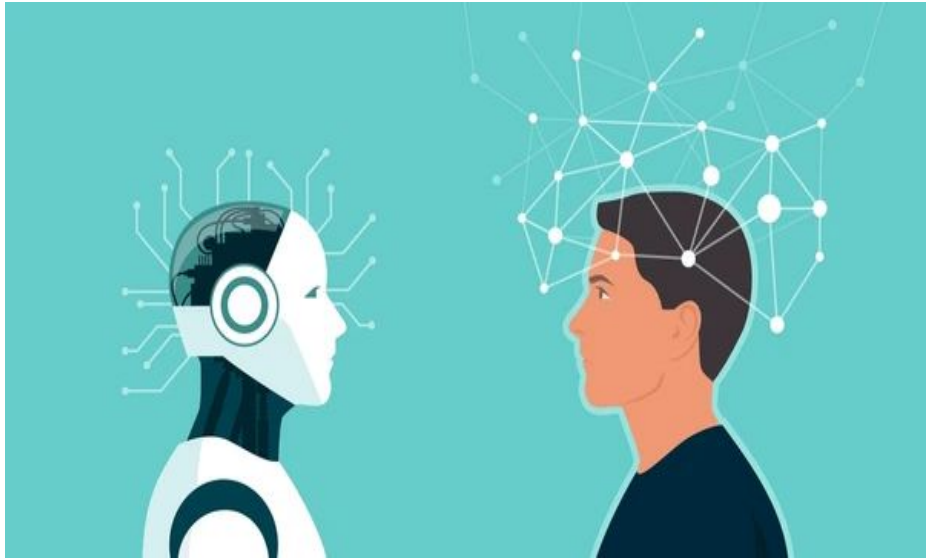
What is Audio Classification?

- **Audio classification** is the task of **automatically identifying and categorizing audio signals** into predefined classes using machine learning or deep learning
- This is a binary audio classification project, in which our classes are Cat and Dog

Why Classify Audio?

- **Classifying audio** is important because it helps **machines understand and interpret sounds**, which enables a wide range of intelligent and interactive applications.
- Humans can naturally tell if a sound is music, speech, or noise. Teaching machines to do the same opens the door to many application



Use cases of Audio Classification

🎙️ 1. Media & Entertainment

- **Music Genre Classification**: Tagging songs for streaming platforms (Spotify, YouTube).

🗣️ 2. Consumer Electronics

- **Contextual Sound Awareness**: Earbuds adjusting noise cancellation based on surrounding sounds (e.g., traffic).

🛡️ 3. Security & Surveillance

- **Anomaly Detection**: Detecting glass breaking, gunshots, or screams in real-time.

🏥 4. Healthcare

Use Cases:

- **Cough and Breath Classification**: Early detection of respiratory conditions like asthma, pneumonia, or COVID.

Project Goal

- Build an AI model that classifies audio clips as either a **cat** or **dog** sound using Deep Learning techniques.
- Differentiating animal sounds helps in **Pet activity monitoring, Smart home automation** (e.g., notify when pet barks/meows), **Animal welfare applications, Voice-enabled pet interaction systems**
- Input: Cat meows, purrs or Dog barks, growls
- Output: **Label**: `'Cat'` or `'Dog'` or **Confidence score** (e.g., 92% dog)

Tools/Stack

- AWS
- Python
- Librosa
- PyTorch
- Flask
- Docker

Thanks

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